1. **MLR has a R-Score Value of 0.789**

```
[18]: from sklearn.linear_model import LinearRegression
      regressor=LinearRegression()
      regressor.fit(X_train, Y_train)

[18]: LinearRegression
      ▶ Parameters

[19]: weight=regressor.coef_
      weight

[19]: array([[  257.8006705 ,   321.06004271,   469.58113407,   -41.74825718,
          23418.6671912 ]])

[20]: bias=regressor.intercept_
      bias

[20]: array([-12057.244846])

[21]: y_pred = regressor.predict(X_test)

[23]: from sklearn.metrics import r2_score
      r_score=r2_score(Y_test, y_pred)
      r_score

[23]: 0.7894790349867009
```

2. **Support Vector Machine:**

S. No	Hyper Parameter	Linear (r score value)	RBF (Non Linear) (r score value)	Poly (r score value)	SIGMOID (r score value)
1	C10	0.462	-0.032	0.038	0.039
2	C100	0.628	0.320	0.617	0.527
3	C500	0.763	0.664	0.826	0.444
4	C1000	0.764	0.810	0.856	0.287
5	C2000	0.744	0.854	0.860	-0.593
6	C3000	0.741	0.866	0.859	-2.124

3. **Decision Tree:**

S. No	Criterion	Max_features	Splitter	R Value
1	squared_error	sqrt	Random	0.701
2	squared_error	sqrt	Best	0.615
3	squared_error	log2	Random	0.683
4	squared_error	log2	Best	0.632
5	friedman_mse	sqrt	Random	0.675
6	friedman_mse	sqrt	Best	0.756
7	friedman_mse	log2	Random	0.735
8	friedman_mse	log2	Best	0.730
9	absolute_error	sqrt	Random	0.688
10	absolute_error	sqrt	Best	0.743
11	absolute_error	log2	Random	0.658
12	absolute_error	log2	Best	0.660
13	poisson	Sqrt	Random	0.682
14	poisson	Sqrt	Best	0.597
15	poisson	log2	Random	0.681
16	poisson	log2	Best	0.746
17	-	-	Random	0.699
18	-	-	Best	0.694

4. **Random Forest:**

S.No	criterion	max_features	n_estimators	R Value
1	squared_error	Sqrt	10	0.852
2	squared_error	Sqrt	50	0.869
3	squared_error	Sqrt	100	0.871
4	squared_error	log2	10	0.852
5	squared_error	log2	50	0.869
6	squared_error	log2	100	0.871
7	friedman_mse	Sqrt	10	0.850
8	friedman_mse	Sqrt	50	0.870
9	friedman_mse	Sqrt	100	0.871
10	friedman_mse	log2	10	0.850
11	friedman_mse	log2	50	0.870
12	friedman_mse	log2	100	0.871
13	absolute_error	Sqrt	10	0.857
14	absolute_error	Sqrt	50	0.870
15	absolute_error	Sqrt	100	0.871
16	absolute_error	log2	10	0.857
17	absolute_error	log2	50	0.870
18	absolute_error	log2	100	0.871
19	poisson	Sqrt	10	0.854
20	Poisson	Sqrt	50	0.863

21	Poisson	Sqrt	100	0.868
22	Poisson	log2	10	0.854
23	Poisson	log2	50	0.863
24	Poisson	log2	100	0.868
25	-	-	10	0.833
26	-	-	50	0.849
27	-	-	100	0.853